Math 303

Final Exam Practice Problems Solutions

Section 1-2B

6. **(a)** Note that 5 toothpicks are added to form each succeeding hexagon. This is an arithmetic sequence with $a_1 = 6$ and $d = 5$, so $a_{10} = 6 + (10 - 1) \cdot 5 = 6 + 9 \cdot 5 =$ **51 toothpicks**.

(b) n hexagons would require $6 + (n - 1) \cdot 5 = 6 + 5n - 5 =$ **$5n + 1$ toothpicks**.

7. **(a)** Looking at the third figure, there are $5 + 3 + 1 = 9$ triangles. The fourth figure would than have $7 + 5 + 3 + 1 = 16$ triangles. An alternative to simply adding 7, 5, 3, and 1 together is to note that $7 + 1 = 8$ and $5 + 3 = 8$. There are $\frac{4}{2} = 2$ of these sums, and $2 \cdot 8 = 16$.

Then the 100th figure would have $100 + 99 = 199$ triangles in the base, $99 + 98 = 197$ triangles in the second row, and so on until the 100th row where there would be 1 triangle. $199 + 1 = 200$; $197 + 3 = 200$; etc. and so the sum of each pair is 200 and there are $\frac{100}{2} = 50$ of these pairs. $50 \cdot 200 = 10{,}000$, or **10,000 triangles** in the 100th figure.

(b) The number of triangles in the nth figure is $\frac{n}{2}$ (number of triangles in base + 1). The number of triangles in the base is $n + (n - 1)$, or $2n - 1$. $(2n - 1) + 1 = 2n$. Then $\frac{n}{2}(2n) = n^2$, or **n^2 triangles** in the nth figure.

8. This is a geometric sequence with $a_1 = 15{,}360$ and $r = \frac{1}{2}$. The nth term of a geometric sequence is $a_n = a_1 r^{n-1}$; thus the 10th term would be $15360\left(\frac{1}{2}\right)^{10} =$ **15 liters**.

Note the progression of terms in the following table:

After Day	Amount of Water Remaining
1	$15{,}360 \cdot \frac{1}{2} = 7680$ liters
2	$7680 \cdot \frac{1}{2} = 3840$ liters
⋮	⋮
9	$60 \cdot \frac{1}{2} = 30$ liters
10	$30 \cdot \frac{1}{2} = 15$ liters

15. **(a)** The first figure has 2 tiles, the second has 5 tiles, the third has 8 tiles, ... This is an arithmetic sequence where the nth term is $2 + (n-1) \cdot 3$.

Thus the 6th term has $2 + (6-1) \cdot 3 =$ **17 tiles**.

(b) The nth term is $2 + (n-1) \cdot 3 = 2 + 3n - 3 =$ $\boldsymbol{3n - 1}$.

(c) The question can be written as: Is there an n such that $3n - 1 = 449$. Since $3n - 1 = 449 \Rightarrow 3n = 450 \Rightarrow n = 150$, the answer is yes, the **150th figure**.

16. The nth term of the arithmetic sequence is $-100 + n(300)$. The sequence can also be generated by adding 300 to the previous term. The nth term of the geometric sequence is 3^{n-1}. The sequence can also be generated by multiplying the previous term by 3. Make a table

Number of the term	Arithmetic term	Geometric term
7	2000	729
8	2300	2187
9	2600	6561

With the **9th term**, the geometric sequence is greater.

Section 2-1B

10. The row of blocks contains one set of 27 blocks, or $1 \cdot 3^3$, plus two sets of 9 each, or $2 \cdot 3^2$ blocks, plus two sets of 3 each, or $2 \cdot 3^1$ blocks, plus two single blocks, or $2 \cdot 3^0$. The representation is thus $\mathbf{1222}_{three}$.

11. **(a)** Place values represent powers of 3; e.g.,

$$1 \cdot 3^0 = 1_{three}$$
$$1 \cdot 3^1 + 0 \cdot 3^0 = 10_{three}$$
$$1 \cdot 3^2 + 0 \cdot 3^1 + 0 \cdot 3^0 = 100_{three}, \text{etc.}$$

Thus the first 10 counting numbers are: $\mathbf{(1,2,10,11,12,20,21,22,100,101)}_{three}$.

(b) Place values represent powers of 8; e.g.,

$$1 \cdot 8^0 = 1_{eight}$$
$$1 \cdot 8^1 + 0 \cdot 8^0 = 10_{eight}$$
$$1 \cdot 8^2 + 0 \cdot 8^1 + 0 \cdot 8^0 = 100_{eight}, \text{etc.}$$

Thus the first 10 counting numbers are: $\mathbf{(1,2,3,4,5,6,7,10,11,12)}_{eight}$.

12. 18 digits. One digit is needed for each of the units from 1 to 17, and one digit for 0.

13. $2022_{three} = (\mathbf{2 \cdot 10^3 + 0 \cdot 10^2 + 2 \cdot 10^1 + 2 \times 10^0})_{three}$. 2022_{three} may alternatively be expanded as $(\mathbf{2 \cdot 3^3 + 0 \cdot 3^2 + 2 \cdot 3^1 + 2 \cdot 3^0})_{ten} = \mathbf{62}$.

14. (a) $\mathbf{222}_{three}$. 2 is the largest units digit in base three.

(b) $\mathbf{EEE}_{twelve}$. E is the largest units digit in base twelve.

15. (a) (*i*) $(100 - 1)_{seven} = 1 \cdot 7^2 + 0 \cdot 7 + 0 - 1 = 6 \cdot 7 + 6 = \mathbf{66}_{seven}$.

(*ii*) $(100 + 1)_{seven} = 1 \cdot 7^2 + 0 \cdot 7 + 0 + 1 = \mathbf{101}_{seven}$.

(b) (*i*) $(10000 - 1)_{two} = 1 \cdot 2^4 + 0 \cdot 2^3 + 0 \cdot 2^2 + 0 \cdot 2^1 + 0 \cdot 2^0 - 1 = 1 \cdot 2^3 + 1 \cdot 2^2 + 1 \cdot 2^1 + 1 \cdot 2^0 = \mathbf{1111}_{two}$.

(*ii*) $(10000 + 1)_{two} = 1 \cdot 2^4 + 0 \cdot 2^3 + 0 \cdot 2^2 + 0 \cdot 2^1 + 0 \cdot 2^0 + 1 = \mathbf{10001}_{two}$.

(c) (*i*) $(101 - 1)_{two} = 1 \cdot 2^2 + 0 \cdot 2^1 + 1 \cdot 2^0 - 1 = 1 \cdot 2^2 + 0 \cdot 2^1 + 0 \cdot 2^0 = \mathbf{100}_{two}$.

(*ii*) $(101 + 1)_{two} = 1 \cdot 2^2 + 0 \cdot 2^1 + 1 \cdot 2^0 + 1 = \mathbf{110}_{two}$.

16. (a) There is **no numeral 6** in base four. The numerals are 0, 1, 2, and 3.

(b) There are **no numerals 2 or 3** in base two. The numerals are 0 and 1.

17. There are 2 groups of 3^3, 2 groups of 3^2, 2 groups of 3^1, and 1 group of 3^0 in 79, so $79 = 2221_{three}$. Thus, where a block is $3 \times 3 \times 3$, a flat is 3×3, and a long is 1×3, **two** blocks, **two** flats, **two** longs, and **one** unit, so, **seven** is the fewest number of base three blocks.

Assessment 2-2B

1. (a) Either a list or set-builder notation may be used: $\{g, e, o, m, t, r, y\}$ or $\{x \mid x$ **is a letter in the word** *geometry*$\}$

(b) $\{8, 9, 10, 11, ...\}$ or $\{x \mid x$ **is a natural number and x>7**$\}$ or $\{x \mid x \in N$ **and x>7**$\}$.

2. (a) $Q = \{a, b, c\}$

(b) $\{1, 3\} = \{3, 1\}$. The symbol $\subset$ refers to a proper subset.

(c) $\{1, 3\} \not\subseteq \{1, 4, 6, 7\}$. The symbol $\subseteq$ refers to a subset.

(d) $0 \notin \varnothing$ **or** $0 \notin \{\,\}$

3. (a) **Yes.** $\{1, 2, 3, 4\} \sim \{w, c, y, z\}$ because both sets have the same number of elements and thus exhibit a one-to-one correspondence.

(b) **Yes**, because both sets have the same number of elements.

(c) **No.** $\{x \mid x$ is a letter in the word *geometry*$\}$ $\neq \{1, 2, 3, 4, ..., 8\}$; there are only seven unduplicated letters in geometry.

15. **(a)** $\subseteq$. Any set is a subset of itself.

(b) $\subseteq$. $\{2\}$ is actually a proper subset of $\{3,2,1\}$.

(c) $\not\subseteq$. 1022 is an element, not a subset.

(d) $\not\subseteq$. 3004 is an element, not a subset.

16. **(a)** **No**. For example, if $A = \{1\}$ and $B = \{1,2\}$, then $A \subseteq B$ but $A \neq B$.

(b) **No**. $A \subset B$ implies that A must have at least one less element than B.

(c) **No**. A and B must have the same number of elements, but not necessarily be equal; for example, if $A = \{1,2\}$ and $B = \{a,b\}$.

(d) **No**. Sec part (c).

Section 3-2B

5. **(a)**

$$\begin{array}{rcrcr} 86 & \Rightarrow & 86+2 & \Rightarrow & 88 \\ -\ 38 & & -\ 38+2 & & -\ 40 \\ \hline & & & & \mathbf{48} \end{array}$$

(b)

$$\begin{array}{rcrcr} 582 & \Rightarrow & 582+6 & \Rightarrow & 588 \\ -\ 44 & & -\ 44+6 & & -\ 50 \\ \hline & & & & \mathbf{538} \end{array}$$

7. Answers may vary, for example:

(a) A tens digit was not regrouped as 1 ten when the sum of the units digits was more than 9.

(b) Partial sums are not in the correct place value position.

(c) The units minuend is subtracted from the units subtrahend.

(d) One tens value should have been traded from the 5 in the minuend's ten position.

11. (a)

$$\begin{array}{cccccc} & 2 & 3 & 4 & 5 \\ + & 8 & 8 & 8 & 8 \\ \hline & \boxed{1/0} & \boxed{1/1} & \boxed{1/2} & \boxed{1/3} \\ 1 & 1 & 2 & 3 & 3 \end{array}$$

(b)

$$\begin{array}{cccccc} & 8 & 7 & 1 & 3 \\ + & 4 & 2 & 1 & 4 \\ \hline & \boxed{1/2} & \boxed{0/9} & \boxed{0/2} & \boxed{0/7} \\ 1 & 2 & 9 & 2 & 7 \end{array}$$

12. Remember that in base five the number 11, for example, means $1 \cdot 5 + 1$, whereas in base ten it means $1 \cdot 10 + 1$.

(a)

$$\begin{array}{cccc} & 3 & 13 & \\ & \not{4} & \not{3} & \textit{five} \\ - & 2 & 4 & \textit{five} \\ \hline & 1 & 4 & \textit{five} \end{array}$$

(b)

$$\begin{array}{cccc} & 1 & 1 & \\ & 1 & 4 & 3 \quad \textit{five} \\ + & & 2 & 3 \quad \textit{five} \\ \hline & 2 & 2 & 1 \quad \textit{five} \end{array}$$

(c)

$$\begin{array}{cccc} & 2 & 12 & \\ & \not{3} & \not{2} & \textit{five} \\ - & 2 & 3 & \textit{five} \\ \hline & & 4 & \textit{five} \end{array}$$

(d)

$$\begin{array}{cccc} & 1 & 1 & \\ & 2 & 3 & 2 \quad \textit{five} \\ + & & 4 & 3 \quad \textit{five} \\ \hline & 3 & 3 & 0 \quad \textit{five} \end{array}$$

(e)

$$\begin{array}{ccccc} 1 & 1 & & & \\ & 1 & 1 & 0 & \textit{two} \\ + & 1 & 1 & 1 & \textit{two} \\ \hline 1 & 1 & 0 & 1 & \textit{two} \end{array}$$

(f)

$$\begin{array}{ccccccc} & 1 & & & & & \\ & \not{1}\not{0} & 10 & & & & \\ \not{1} & \not{0} & \not{0} & 0 & 1 & & \textit{two} \\ - & & 1 & 0 & 1 & & \textit{two} \\ \hline & 1 & 1 & 0 & 0 & & \textit{two} \end{array}$$

16. **(a)** Use the scratch algorithm. Whenever a sum is 10 or more, scratch a line through the last digit added and write the number of units next to the scratched digit; count the number of scratches in each column and add to the column to the left.

$$\begin{array}{rcccc} & & 1 & 2 & \\ & & 5 & 3 & 7 \\ & & 3 & 1 & \not{8}_5 \\ + & {}^{1}2 & \not{3}_2 & \not{4}_0 & \not{5}_0 \\ \hline & 3 & 2 & 0 & 0 \end{array}$$

(b) A base six addition table might be helpful (see problem 13):

$$\begin{array}{rcccl} & {}^{2}\not{4}_0 & 1 & _{six} \\ & 3 & 2 & _{six} \\ & 2 & 2 & _{six} \\ & \not{4}_3 & \not{3}_2 & _{six} \\ & 2 & 2 & _{six} \\ + & \not{5}_4 & \not{4}_2 & _{six} \\ \hline 3 & 4 & 2 & _{six} \end{array}$$

17. There is **no numeral 6 in base six**. $23_{six} + 43_{six} = 110_{six}$.

18. **(a)** Subtract 213_{five} from 342_{five}.

$$\begin{array}{rcccl} & 3 & 4 & 2 & _{five} \\ - & 2 & 1 & 3 & _{five} \\ \hline & \mathit{1} & \mathit{2} & \mathit{4} & _{five} \end{array}$$

(b) Subtract 1011_{two} from 1011_{two}.

$$\begin{array}{rcccl} & 1 & 1 & 0 & 1 & _{two} \\ - & 1 & 0 & 1 & 1 & _{two} \\ \hline & & & \mathbf{1} & \mathbf{0} & _{two} \end{array}$$

(c) Subtract 9 from $E08_{twelve}$:

$$\begin{array}{rcccl} & E & 0 & 8 & _{twelve} \\ - & & & 9 & _{twelve} \\ \hline & T & E & E & _{twelve} \end{array}$$

(d) Subtract 100_{two} from $10,000_{two}$.

$$\begin{array}{rccccc} & 1 & 0 & 0 & 0 & 0 & _{two} \\ - & & & 1 & 0 & 0 & _{two} \\ \hline & & \mathbf{1} & \mathbf{1} & \mathbf{0} & \mathbf{0} & \mathbf{_{two}} \end{array}$$

Section 3-3B

2. (a) **No**, $2 \cdot 2 = 4$ which is not in $\{1, 2\}$.

(b) **Yes**, the expression $2k + 1, k \in W$, represents any odd number.

Let $2\ell + 1$ be another odd number, then $(2k + 1)(2\ell + 1) = 4k\ell + 2k + 2\ell = 2(k\ell + k + \ell) + 1$ which is an odd number.

(c) **Yes**, a power of 2 multiplied by a power of 2 is still a power of 2.

10. (a) **Zero multiplication property of whole numbers**; i.e., $a \cdot 0 = 0$.

(b) **Commutative property of multiplication of whole numbers**; i.e., $a \cdot b = b \cdot a$.

(c) **Commutative property of multiplication;** i.e., $a(bc) = (bc)a$.

(d) **Identity property of multiplication of whole numbers**; i.e., $a \cdot 1 = a$.

(e) **Distributive property of multiplication over addition for whole numbers**; i.e. $(a + b)c = ac + bc$.

(f) **Distributive property of multiplication over addition for whole numbers**; i.e., $(a + b)(c + d) = (a + b) \cdot c + (a + b) \cdot d$.

30. (a) **Yes**. $x \odot y \in S$ for any values of $x \in S$ and $y \in S$.

(b) **Yes**. $a \odot b = b \odot a, a \odot c = c \odot a, \ldots$ for all possible combinations of terms.

(c) **Yes**. Since $x \odot a = x$ and $a \odot x = x$ for all x in S, a is the identity on S.

(d) **Yes**. $a \odot (b \odot c) = (a \odot b) \odot c$ for all possible combinations of terms.

Assessment 3-5B

1. Answers may vary; possibilities include:

(a) $160 + 92 - 32 + 40 - 18$

$= 160 + 92 + 40 - 32 - 18$

$= (160 + 90 + 40 + 2) - 32 - 18$

$= (290 + 2) - 32 - 18$

$= 292 - 32 - 18 = 292 - 50 = \mathbf{242}$.

(b) $36 + 97 - 80 + 44$

$= 36 + 97 + 44 - 80$

$= (30 + 90 + 40 + 6 + 7 + 4) - 80$

$= (160 + 17) - 80$

$= 177 - 80 = \mathbf{97}$.

2. Answers may vary; possibilities include:

(a) Rearrange as $(11 \cdot 3) \cdot (5 \cdot 20) = 33 \cdot 100 =$ **3300**.

(b) Rearrange as $(82 + 18) + (37 + 13) =$ $100 + 50 = \mathbf{150}$.

3. Step 1: **Associative** property of addition.

Step 2: **Commutative** property of addition.

Step 3: **Associative** property of addition.

Step 4: **Closure property of whole-number addition**.

4. **(a)** $997 - 32$:

(*i*) $997 - 30 = 967$ (subtract the tens value of 32 from 997).

(*ii*) $967 - 2 = \mathbf{965}$ (subtract the units value of 32 from 967).

(b) $56 \cdot 30$:

(*i*) $50 \cdot 30 = 1500$ (multiply the tens value of 56 by 30).

(*ii*) $6 \cdot 30 = 180$ (multiply the units value of 56 by 30).

(*iii*) $1500 + 180 = \mathbf{1680}$ (add the partial products).

5. **(a)** $75 - 38 \Rightarrow (75 + 2) - (38 + 2) = 77 - 40 = \mathbf{37}$.

(b) $57 + 35 \Rightarrow (57 + 3) + (35 - 3) = 60 + 32 = \mathbf{92}$.

(c) $137 - 29 \Rightarrow (137 + 1) - (29 + 1) = 138 - 30 = \mathbf{108}$.

(d) $78 + 49 \Rightarrow (78 + 2) + (49 - 2) = 80 + 47 = \mathbf{127}$.

6. **(a)** $63 + 7 = 70$; $70 + 4 = 74$. Then $7 + 4 = \mathbf{11}$.

(b) $57 + 3 = 60$; $60 + 10 = 70$; $70 + 3 = 73$. Then $3 + 10 + 3 = \mathbf{16}$.

7. Methods may vary.

(a) $81 - 46 = (81 - 1) - (46 - 1) = 80 - 45 =$ **35** (trading off).

(b) $98 - 19 \Rightarrow 19 + 1 = 20$; $20 + 70 = 90$; $90 + 8 = 98$. Then $1 + 70 + 8 = \mathbf{79}$ (cashier's algorithm).

(c) $9700 - 600 \Rightarrow 97 - 6 = 91 \Rightarrow \mathbf{9100}$ (drop the zeros).

---

8. Mentally multiply $7 \times 70 = 490$ miles and compensate by subtracting 7 miles for an anwser of 483 miles .

7 hrs(69 miles/hr) $= 7(70 - 1)$ miles $=$ **7(70) − 7 miles.**

9. **(a)** $3\underline{5}87$: The number is between 3500 and 3600;
The midpoint is 3550;
The number is greater than the midpoint;
So it rounds up to **3600**.

(b) $1\,48,213$: The number is between 100,000 and 200,000;
The midpoint is 150,000;
The number is less than the midpoint;
So it rounds down to **100,000**.

(c) $2\underline{3},785$: The number is between 23,000 and 24,000;
The midpoint is 23,500;
The number is greater than the midpoint;
So it rounds up to **24,000**.

(d) 2357: The number is between 2350 and 2360;
The midpoint is 2355;
The number is greater than the midpoint;
So it rounds up to **2360**.

10. Estimates may vary.

(a) $937 \div 38 \approx 900 \div 30 = \mathbf{30}$.

(b) $32,285 - 18,988 \approx 32,000 - 19,000 = \mathbf{13,000}$.

(c) $52 \cdot 49 \approx 50 \cdot 50 = \mathbf{2500}$.

(d) $3215 + 3789 + 5987 \approx 3000 + 4000 + 6000 = \mathbf{13,000}$.

11. Estimates may vary.

(a) $2345 + 5250 + 4210 + 910$:

(*i*) $2 + 5 + 4 + 0 = 11$ (add front-end digits);

(*ii*) $11{,}000$ (place value);

(*iii*) $345 + 250 + 210 + 910 \approx 350 + 250 + 200 + 900 = 1700$ (adjust);

(*iv*) $11{,}000 + 1700 = \mathbf{12{,}700}$ (adjusted estimate).

(b) $345 + 518 + 655 + 270$:

(*i*) $3 + 5 + 6 + 2 = 16$ (add front-end digits);

(*ii*) 1600 (place value);

(*iii*) $50 + 20 + 60 + 70 = 200$ (adjust);

(*iv*) $1600 + 200 = 1800$ (adjusted estimate).

Section 4-2B

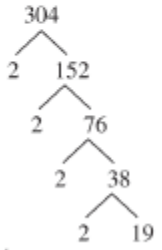
3. **(a)** 2|304; the factor tree starts with $2 \cdot 152 = 304$.

2|152; the next level is $2 \cdot 76 = 152$.

2|76; the next level is $2 \cdot 38 = 76$.

2|76; the final level is $2 \cdot 19 = 38$, since 19 is prime.

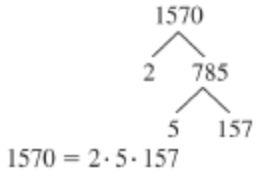
Thus the factor tree is:



$304 = 2^4 \cdot 19$

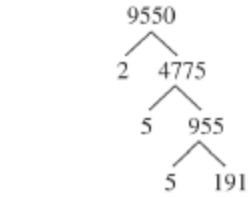
(b)

(b)



$1570 = 2 \cdot 5 \cdot 157$

(c)



$9550 = 2 \cdot 5^2 \cdot 191$

5. **19**. $19^2 < 503$, but $23^2 > 503$. (19 is prime and 23 is the next largest prime.)

7. **(a)** **3, 5, 15, or 29** members. Prime factorization of $435 = 3 \cdot 5 \cdot 29$, so committees could have these numbers of members or products thereof.

(b) **145** three-member committees $(435 \div 3)$, **87** five-member committees, **29** fifteen-member committees, or **15** twenty-nine member committees.

17. (*i*) $2^2 \cdot 5^3 \cdot 9^2$ is not a prime factorization because 9 is not prime.

(*ii*) The prime factorization is $2^2 \cdot 5^3 \cdot (3^2)^2 = \mathbf{2^2 \cdot 5^3 \cdot 3^4}$.

20. If $12|n$ there exists a whole number a such that $12a = n$:

$(3 \cdot 4)a = n$

$3(4a) = n$

Since $4a$ is a whole number, $3|n$.

Assessment 4-3B

1. **(a)** (*i*) $D_{12} = \{1, 2, 3, 4, 6, 12\}$

$D_{18} = \{1, 2, 3, 6, 9, 18\}$

$\Rightarrow GCD(12,18) = \mathbf{6}$.

(*ii*) $M_{12} = \{12, 24, 36, 48, 60, 72, 84, \ldots\}$

$M_{18} = \{18, 36, 54, 72, 90, 108, ...\}$

$\Rightarrow LCM(12,18) = \mathbf{36}.$

(b) (*i*) $D_{18} = \{1, 2, 3, 6, 9, 18\}$

$D_{36} = \{1, 2, 3, 4, 6, 9, 12, 18, 36\}$

$\Rightarrow GCD(18,36) = \mathbf{18}.$

(*ii*) $M_{18} = \{18, 36, 54, 72, 90, 108, \ldots\}$

$M_{36} = \{36, 72, 108, 144, 180, ...\}$

$\Rightarrow LCM(18, 36) = \mathbf{36}.$

(c) (*i*) $D_{12} = \{1, 2, 3, 4, 6, 12\}$

$D_{18} = \{1, 2, 3, 6, 9, 18,\}$

$D_{24} = \{1, 2, 3, 4, 6, 8, 12, 24\}$

$\Rightarrow GCD(12,18, 24) = \mathbf{6}.$

(*ii*) $M_{12} = \{12, 24, 36, 48, 60, 72, 84, ...\}$

$M_{18} = \{18, 36, 54, 72, 90, 108, ...\}$

$m_{24} = \{24, 48, 72, 96, 120, 144, ...\}$

$\Rightarrow LCM(12,18, 24) = \mathbf{72}.$

(d) (*i*) $D_6 = \{1, 2, 3, 6\}$

$D_{11} = \{1, 11\}$

$\Rightarrow GCD(6,11) = \mathbf{1}.$

(*ii*) $M_6 = \{6, 12, 18, 24, 30, ..., 66, 72, ...\}$

$M_{11} = \{11, 22, 33, 44, 55, 66, 77, ...\}$

$\Rightarrow LCM(6, 11) = \mathbf{66}.$

2. **(a)** 11 = prime number

19 = prime number

$\Rightarrow GCD(11, 19) = \mathbf{1}.$

$LCM(11, 19) = 11 \cdot 19 = \mathbf{209}.$

(b) $140 = 2 \cdot 2 \cdot 5 \cdot 7 = 2^2 \cdot 5 \cdot 7$

$320 = 2 \cdot 2 \cdot 2 \cdot 2 \cdot 2 \cdot 2 \cdot 5 = 2^6 \cdot 5$

$\Rightarrow GCD(140, 320) = 2^2 \cdot 5 = \mathbf{20}$

$LCM(140, 320) = 2^6 \cdot 5 \cdot 7 = \mathbf{2240}.$

(c) $800 = 2 \cdot 2 \cdot 2 \cdot 2 \cdot 2 \cdot 5 \cdot 5 = 2^5 \cdot 5^2$

$75 = 3 \cdot 5 \cdot 5 = 3 \cdot 5^2$

$450 = 2 \cdot 3 \cdot 3 \cdot 5 \cdot 5 = 2 \cdot 3^2 \cdot 5^2$

$\Rightarrow GCD(800, 75, 450) = 5^2 = \mathbf{25}$

$LCM(800, 75, 450) =$

$2^5 \cdot 3^2 \cdot 5^2 = \mathbf{7200}.$

(d) $104 = 2 \cdot 2 \cdot 2 \cdot 13 = 2^3 \cdot 13$

$320 = 2 \cdot 2 \cdot 2 \cdot 2 \cdot 2 \cdot 2 \cdot 5 = 2^6 \cdot 5$

$\Rightarrow GCD(104, 320) = 2^3 = \mathbf{8}$

$LCM(104, 320) = 2^6 \cdot 5 \cdot 13 = \mathbf{4160}.$

3. $\rightarrow R$ symbolizes the reminder left after the indicated divisions.

(a) $GCD(14560, 8250) = GCD(8250, 6310)$
since $14560 \div 8250 \rightarrow R\,6310$

$GCD(8250, 6310) = GCD(6310, 1940)$
since $8250 \div 6310 \rightarrow R\,1940$

$GCD(6310, 1940) = GCD(1940, 490)$ since
$6310 \div 1940 \rightarrow R\,490$

$GCD(1940, 490) = GCD(490, 470)$ since
$1940 \div 490 \rightarrow R\,470$

$GCD(490, 470) = GCD(470, 20)$ since
$490 \div 470 \rightarrow R\,20$

$GCD(470, 20) = GCD(20, 10)$ since
$470 \div 20 \rightarrow R\,10$

$GCD(20, 10) = GCD(10, 0)$ since
$20 \div 10 \rightarrow R\,0$

$\Rightarrow GCD(14560, 8250) = \mathbf{10}.$

(b) $GCD(8424, 2520) = GCD(2520, 864)$
since $8424 \div 2520 \rightarrow R\,864$

$GCD(2520, 864) = GCD(864, 792)$ since
$2520 \div 864 \rightarrow R\,792$

$GCD(864, 792) = GCD(792, 72)$ since
$864 \div 792 \rightarrow R\,72$

$GCD(792, 72) = GCD(72, 0)$ since
$792 \div 72 \rightarrow R\,0$

$\Rightarrow GCD(8424, 2520) = \mathbf{72}.$

4. **(a)** (*i*) Intersection of sets:

$M_{25} = \{25, 50, 75, 100, ..., 900, ...\}$

$M_{36} = \{36, 72, 108, 144, ..., 900, ...\}$

$\Rightarrow LCM(25, 36) = \mathbf{900}.$

(*ii*) Prime factorization:

$25 = 5 \cdot 5 = 5^2$

$36 = 2 \cdot 2 \cdot 3 \cdot 3 = 2^2 \cdot 3^2$

$\Rightarrow LCM(25, 36)$

$= 2^2 \cdot 3^2 \cdot 5^2 = \mathbf{900}.$

(b) Prime factorization:

$82 = 2 \cdot 41$

$90 = 2 \cdot 3 \cdot 3 \cdot 5 = 2 \cdot 3^2 \cdot 5$

$50 = 2 \cdot 5 \cdot 5 = 2 \cdot 5^2$

$\Rightarrow LCM(82, 90, 50)$

$= 2 \cdot 3^2 \cdot 5^2 \cdot 1 = \mathbf{18,450}.$

(c) Prime factorization:

$80 = 2 \cdot 2 \cdot 2 \cdot 2 \cdot 5 = 2^4 \cdot 5$

$105 = 3 \cdot 5 \cdot 7$

$315 = 3 \cdot 3 \cdot 5 \cdot 7 = 3^2 \cdot 5 \cdot 7$

$\Rightarrow LCM(80, 105, 315)$

$= 2^4 \cdot 3^2 \cdot 5 \cdot 7 = \mathbf{5040}.$

(d) Prime factorization:

$8^{100} = (2^3)^{100} = 2^{300}$

$50^{100} = (2 \cdot 5^2)^{100} = 2^{100} \cdot 5^{200}$

$\Rightarrow LCM(8^{100}, 50^{100}) = \mathbf{2^{300} \cdot 5^{200}}.$

9. The question is really "What is $LCM\ (6, 8)$?" since that is when the off nights will coincide:
$LCM\ (6, 8) =$ **24 nights**.

10. $LCM\ (90, 75) = 450$ minutes, or $7\frac{1}{2}$ hours;

7:00 P.M. $+ 7\frac{1}{2}$ hours $=$ **2:30 A.M.**

Section 5-1B

3. **(a)** Absolute value is the distance on a number line between the origin (0) and a specified number. Distance on the number line between 0 and $^-3$ is 3 units, so $|^-3| = \mathbf{3}$.

(b) Distance on the number line between 0 and 15 is 15 units, so $|15| = \mathbf{15}$.

(c) $^-|^-3|$ means the additive inverse of the absolute value of $^-3$, so $^-|^-3| = ^-(3) = \mathbf{^-3}$.

(d) $^-|6| = ^-6$

4. **(a)** Charged-field model:

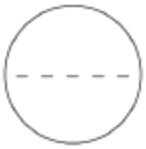


⁻2 charge



Put in 5 positive charges. Net result: 3 positive charges; answer: 3

(b) Charged-field model:



⁻5 charge

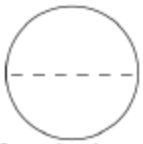


Put in 2 positive charges. Net result: 3 negative charges; answer: ⁻3

(c) Charged-field model:



⁻3 charge



Put in 3 negative charges: Net result: 6 negative charges; answer: ⁻6

9. **(a)** $^{-}247 + {}^{-}11 + {}^{-}11 = {}^{-}258 + {}^{-}11$ $= \mathbf{{}^{-}269^{\circ}\ C}$.

(b) $98 - {}^{-}94 = 98 + {}^{-}({}^{-}94) = 98 + 94 =$ **192°F**.

20. **(a)** There are **no values** of x for which this is true.

If $^{-}|x| = 2$, then $|x| = {}^{-}2$, which is not possible.

(b) **All integers except 0.** Since $|x|$ is always non-negative, $^{-}|x|$ is always negative for all integers except 0; 0 is neither negative nor positive.

(c) There are **no values** of x for which this is true.

$^{-}|x|$ is always negative for all integers except 0, which is neither negative nor positive.

(d) $^{-}x + 1$ is positive means $-x + 1 > 0$.

$-x + 1 > 0 \Rightarrow -x > -1 \Rightarrow x < \mathbf{1}$.

(e) $^{-}x - 1$ is negative means $-x - 1 < 0$.

$-x - 1 < 0 \Rightarrow -x < 1 \Rightarrow x > \mathbf{-1}$.

21. If $y = |x - 5|$:

(a) $y = |(10) - 5| = |10 + {}^{-}5| = |5| = \mathbf{5}$, when $x = 10$.

(b) $f({}^{-}1) = |({}^{-}1) - 5| = |{}^{-}1 + {}^{-}5| = |{}^{-}6| = \mathbf{6}$, when $x = {}^{-}1$.

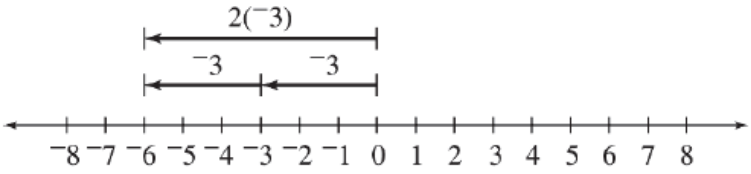
(c) If $7 = |x - 5| \Rightarrow x - 5 = 7$ or $x - 5 = {}^{-}7$:

(i) $x - 5 = 7 \Rightarrow x = 7 + 5 \Rightarrow x = \mathbf{12}$.

(ii) $x - 5 = {}^{-}7 \Rightarrow x = {}^{-}7 + 5 \Rightarrow x = {}^{-}\mathbf{2}$.

Section 5-2B

3. Move three units to the left twice to arrive at the product $2(^-3) = {}^-6$:



4. **(a)** Take away two groups of three negative charges, leaving six positive charges; i.e., $\mathbf{(^-2) \cdot (^-3) = 6}$.

(b) Take away four groups of two positive charges, leaving eight negative charges; i.e., $\mathbf{(^-4) \cdot 2 = {}^-8}$.

8. For each of the following, if $a \cdot b = c \Rightarrow c \div b = a$ and $c \div a = b$ $(a \neq 0$ and $b \neq 0)$:

(a) ***(i)*** $^-5 \cdot 4 = {}^-20$

(ii) $^-20 \div 4 = {}^-5$

(iii) $^-20 \div {}^-5 = 4$.

(b) ***(i)*** $(^-4) \cdot (^-3) = 12$

(ii) $12 \div (^-4) = {}^-3$

(iii) $12 \div (^-3) = {}^-4$.

(c) $0 \div 0$ is not defined. To understand why suppose there is one answer and name it x. If $0 \div 0 = x$, then $0 \cdot x = 0$. Notice that any real numbers could replace x and $0 \cdot x = 0$ would be true. For example, $0 \cdot 0 = 0$ and $0 \cdot 13 = 0$. Since the solution to $0 \cdot x = 0$ is not unique, $0 \div 0$ is not defined.

10. **(a)** $^{-}5^{\circ}\text{C} + (m \text{ minutes} \times d^{\circ} \text{ per minute}) =$
$^{-}5^{\circ} + md^{\circ} = \mathbf{(^{-}5 + md)^{\circ}\ C}.$

(b) $0^{\circ}\text{C} + (^{-}m \text{ minutes} \times {}^{-}d^{\circ} \text{ per minute}) =$
$0^{\circ} + md^{\circ} = \boldsymbol{md^{\circ}}\mathbf{C}.$

(c) $20^{\circ}\text{C} + (^{-}m \text{ minutes} \times d^{\circ} \text{ per minute}) =$
$20^{\circ} + {}^{-}md^{\circ} = \mathbf{(20 - md)^{\circ}\ C}.$

28. **(a)** **False**. Let $x = 3$ and $y = {}^{-}5$.
Then $|x + y| = |3 + {}^{-}5| = |{}^{-}2| = 2$
but $|x| + |y| = |3| + |{}^{-}5| = 3 + 5 = 8.$

(b) **True**. If $x \geq 0$ and $y \geq 0$, then $xy \geq 0$ so $|xy| = xy$ and $|x| \cdot |y| = xy$. If $x \leq 0$ and $y \geq 0$, then $xy \leq 0$ so $|xy| = {}^{-}(xy)$ and $|x| \cdot |y| = {}^{-}x \cdot y = {}^{-}(xy)$; similarly for $x \geq 0$ and $y \leq 0$. If $x \leq 0$ and $y \leq 0$ then $xy \geq 0$ so $|xy| = xy$ and $|x| \cdot |y| = ({}^{-}x) \cdot ({}^{-}y) = xy.$

(c) **True**. Because $x^2 \geq 0, |x^2| = x^2.$

(d) **True**. From parts (b) and (c);
$|x|^2 = |x| \cdot |x| = |x \cdot x| = |x^2| = x^2.$

Assessment 6-1B

1. (a) The solution to $10x = 7$ is $\frac{7}{10}$.

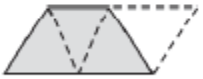
   (b) Joe ate seven of the ten apple slices.

   (c) The ratio of boys to girls in this math class is seven to ten.

5. (a) Divide the figure into three equal parts, then add another equal part. The shaded area is then the whole:



(b) Divide the figure into four equal parts, then remove one of the equal parts. The shaded area is then the whole:



(c) If two circles are $\frac{1}{5}$ or $\frac{2}{10}$ of the whole, then the whole is ten shaded circles:



(d) If the rectangle is $\frac{1}{4}$ of the whole, then add three other equal rectangles:



7. Answers may vary.

(a) $\frac{2}{6}, \frac{3}{9}, \frac{4}{12}$.

(b) $\frac{8}{10}, \frac{12}{15}, \frac{16}{20}$.

(c) $\frac{-6}{14}, \frac{-9}{21}, \frac{-12}{28}$.

(d) $\frac{2a}{6}, \frac{3a}{9}, \frac{4a}{12}$.

11. **(a)** $\frac{6}{16} = \frac{6}{4^2} = \frac{6}{4^2} \cdot \frac{25}{25} \cdot \frac{25}{25} = \frac{6 \cdot 25 \cdot 25}{4 \cdot 25 \cdot 4 \cdot 25} = \frac{3{,}750}{10{,}000}$. The fractions are **equal**.

(b) **Not equal**. $\frac{17}{27}$ is in its reduced form and $\frac{25}{45} = \frac{5 \cdot 5}{5 \cdot 9} = \frac{5}{9}$ or $17 \cdot 45 \neq 27 \cdot 25$.

17. **(a)** $\frac{1}{-7} < \frac{1}{-8}$. $LCD(7, 8) = 56 \Rightarrow$

$\frac{1}{-7} = \frac{-1}{7} = \frac{-8}{56}$ and $\frac{1}{-8} = \frac{-1}{8} = \frac{-7}{56}$,

or $1 \cdot {}^-8 < {}^-7 \cdot 1$.

(b) $\frac{2}{5} = \frac{4}{10} \cdot 2 \cdot 10 = 5 \cdot 4$.

(c) $\frac{0}{7} = \frac{0}{17} \cdot \frac{0}{7} = 0 = \frac{0}{17}$.

21. $\dfrac{\text{Number of miles} \mid 120 \mid x}{\text{Number of inches} \mid 1 \mid 3 \,/\, 4} \Rightarrow \frac{120 \text{ miles}}{1 \text{ inch}} = \frac{x \text{ miles}}{3/4 \text{ inches}} \Rightarrow 120 \cdot 3/4 = 1 \cdot x \Rightarrow \mathbf{90 \text{ miles}} = x.$

Assessment 6-2B

1. Various methods may be used:

(a) $\frac{^-1}{2} + \frac{2}{3} \Rightarrow \text{LCD}(2,3) = 6 \Rightarrow \frac{^-1}{2} \cdot \frac{3}{3} + \frac{2}{3} \cdot \frac{2}{2} = \frac{^-3}{6} + \frac{4}{6} = \mathbf{\frac{1}{6}}$.

(b) $\frac{5}{12} - \frac{2}{3} = \frac{5 \cdot 3 - 2 \cdot 12}{12 \cdot 3} = \frac{15 - 24}{36} = \frac{^-9}{36} = \mathbf{\frac{^-1}{4}}$.

(c) $\frac{5}{4x} + \frac{^-3}{2y} = \frac{5 \cdot 2y + {}^-3 \cdot 4x}{4x \cdot 2y} = \frac{10y - 12x}{8xy} = \mathbf{\frac{5y - 6x}{4xy}}$.

(d) $\frac{^-3}{2x^2y^2} + \frac{5}{2xy^2} + \frac{7}{x^2y} \Rightarrow$ $\text{LCD}(2x^2y^2, 2xy^2, x^2y) = 2x^2y^2 \Rightarrow \frac{-3}{2x^2y^2} + \frac{5}{2xy^2} \cdot \frac{x}{x} + \frac{7}{x^2y} \cdot \frac{2y}{2y} = \mathbf{\frac{^-3 + 5x + 14y}{2x^2y^2}}$.

(e) $\frac{5}{6} - 2\frac{1}{8} = \frac{5}{6} - \frac{17}{8} = \frac{5 \cdot 8 - 6 \cdot 17}{6 \cdot 8} = \frac{40 - 102}{48} = \frac{^-62}{48} = \mathbf{\frac{^-31}{24}} = \mathbf{^-1\frac{7}{24}}$.

(f) $^-4\frac{1}{2} + 3\frac{1}{6} \Rightarrow \text{LCD}(2,6) = 6 \Rightarrow$ $^-4\frac{1}{2} \cdot \frac{3}{3} + 3\frac{1}{6} = {}^-4\frac{3}{6} + 3\frac{1}{6} = {}^-1\frac{2}{6} = \mathbf{^-1\frac{1}{3}} = \mathbf{\frac{^-4}{3}}$.

8. **(a) 1; low.** $\frac{113}{100} > 1$.

(b) **0; low.** $\frac{3}{1978} > 0$.

(c) $\mathbf{\frac{3}{4}}$**; low.** $\frac{150}{198} > \frac{150}{200} = \frac{3}{4}$.

(d) **1; high.** $\frac{8}{9} < 1$.

14. $3\frac{1}{2} - 1\frac{3}{4} = 3\frac{2}{4} - 1\frac{3}{4} = \left(2\frac{2}{4} + \frac{4}{4}\right) - 1\frac{3}{4} = 2\frac{6}{4} - 1\frac{3}{4} = \mathbf{1\frac{3}{4}}$ **cups.**

15. $38\frac{1}{4} - \left(15\frac{3}{4} + \frac{3}{8}\right) = 38\frac{2}{8} - \left(15\frac{6}{8} + \frac{3}{8}\right) = 38\frac{2}{8} - 16\frac{1}{8} = \mathbf{22\frac{1}{8}}$ **inches.**

19. (a) $x - \frac{5}{6} = \frac{2}{3} \Rightarrow x = \frac{5}{6} + \frac{2}{3} = \frac{5}{6} + \frac{4}{6} = \frac{9}{6} = \frac{3}{2}$. $x = \mathbf{1\frac{1}{2}}$.

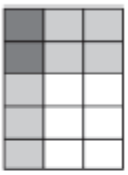
(b) $x - \frac{7}{2^3 \cdot 3^2} = \frac{5}{2^2 \cdot 3^2} \Rightarrow x = \frac{5}{2^2 \cdot 3^2} + \frac{7}{2^3 \cdot 3^2} = \frac{5}{2^2 \cdot 3^2} \cdot \frac{2}{2} + \frac{7}{2^3 \cdot 3^2} = \frac{10+7}{2^3 \cdot 3^2} \Rightarrow x = \frac{\mathbf{17}}{\mathbf{2^3 \cdot 3^2}} = \frac{\mathbf{17}}{\mathbf{72}}$.

21. Write the equivalent operations of using (take away) paint and adding paint:

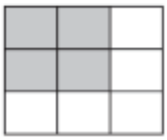
$$\frac{3}{4} - \frac{1}{3} + \frac{1}{2} = \frac{9}{12} - \frac{4}{12} + \frac{6}{12} = \frac{\mathbf{11}}{\mathbf{12}} \textbf{ cup}.$$

Section 6-3B

2. (a) There are 15 boxes in the figure below. The dark-shaded region represents $\frac{6}{15}$ of $\frac{5}{15}$, or $\frac{2}{5} \cdot \frac{1}{3} = \frac{\mathbf{2}}{\mathbf{15}}$:



(b) There are 9 boxes in the figure below. The dark-shaded region represents $\frac{6}{9}$ of $\frac{6}{9}$, or $\frac{2}{3} \cdot \frac{2}{3} = \frac{4}{9}$.



5. (a) $\frac{\mathbf{7}}{\mathbf{6}}$. $\frac{6}{7} \cdot \frac{7}{6} = 1$.

(b) $\frac{\mathbf{1}}{\mathbf{8}}$. $8 \cdot \frac{1}{8} = 1$.

(c) $4\frac{1}{5} = \frac{21}{5}$. $\frac{21}{5} \cdot \frac{\mathbf{5}}{\mathbf{21}} = 1$.

(d) $^{-}1\frac{1}{2} = \frac{-3}{2}$. $\frac{-3}{2} \cdot \left(\frac{\mathbf{-2}}{\mathbf{3}}\right) = 1$.

10. (a) **Greater than 2**. If the quotient was 2, then checking would show that $2\left(2\frac{13}{100}\right) = 4\frac{26}{100}$, which is less than $4\frac{1}{3}$. Thus the true quotient must be greater than 2.

(b) **Less than 4**. $16 \div 4\frac{3}{18}$ is 16 divided by a number more than 4.

(c) **Greater than 4**. $16 \div 3\frac{8}{9}$ is 16 divided by a number less than 4.

14. (a) Increasing a salary by $\frac{1}{10}$ means the salary will be $1\frac{1}{10} = \frac{11}{10}$ of its previous value. With two such raises Martha will make $\left(100{,}000 \cdot \frac{11}{10}\right) \cdot \frac{11}{10} = 110{,}000 \cdot \frac{11}{10} =$ **$121,000**.

(b) $99,000 is $\frac{11}{10}$ of **what Aaron made last year**, i.e., $99{,}000 = \frac{11}{10}s$ (where s is last year's salary) $\Rightarrow s = \frac{10}{11}(99{,}000) =$ **$90,000**.

(c) Let s be Juanita's salary two years ago. Then $363{,}000 = \frac{11}{10}\left(\frac{11}{10}s\right)$, since two raises brought her to that point. $\frac{121}{100}s = 363{,}000 \Rightarrow s = \frac{100}{121}(363{,}000) =$ **$300,000**.

Section 6-4B

2. $\frac{a}{b} = \frac{c}{d}$ only if $ad = bc$:

(a) $\frac{5}{x} = \frac{30}{42} \Rightarrow 30x = 5 \cdot 42 \Rightarrow x = \frac{5 \cdot 42}{30} = \frac{210}{30} =$ **7**.

(b) $\frac{x}{8} = \frac{-12}{32} \Rightarrow 32x = {}^{-}12 \cdot 8 \Rightarrow x = \frac{{}^{-}12 \cdot 8}{32} =$ **$^{-}$3**.

(c) $\frac{7}{8} = \frac{3x}{48} \Rightarrow 3x \cdot 8 = 7 \cdot 48 \Rightarrow x = \frac{7 \cdot 48}{3 \cdot 8} =$ **14**.

(d) $3\frac{1}{2}$ is to 8 as x is to 24 $\Rightarrow \frac{3\frac{1}{2}}{8} = \frac{x}{24} \Rightarrow 8x = 3\frac{1}{2} \cdot 24 \Rightarrow x = \frac{3\frac{1}{2} \cdot 24}{8} =$ **$10\frac{1}{2}$**.

7. $\frac{9 \text{ months}}{6 \text{ vocation days}} = \frac{12 \text{ months}}{x \text{ vocation days}} \Rightarrow 9 \cdot x = 6 \cdot 12 \Rightarrow x = \frac{6 \cdot 12}{9} = \mathbf{8 \text{ days}}$.

8. $\frac{1 \text{ teacher}}{30 \text{ students}} = \frac{x \text{ teachers}}{1200 \text{ students}} \Rightarrow 30x = 1200 \Rightarrow x = 40$ teachers presently.

Then $\frac{1 \text{ teacher}}{20 \text{ students}} = \frac{40 + t \text{ additional teachers}}{1200 \text{ students}} \Rightarrow 20(40 + t) = 1200 \Rightarrow 800 + 20t = 1200 \Rightarrow 20t = 400 \Rightarrow t = \frac{400}{20} = \mathbf{20 \text{ additional teachers}}$.

14. The ratio of length to width of the park is $\frac{500 \text{ ft}}{300 \text{ ft}} = \frac{5}{3}$. Then $\frac{5 \text{ in}}{3 \text{ in}} = \frac{4 \text{ in}}{x \text{ in}} \Rightarrow 5x = 4.3 \Rightarrow x = \frac{4(3)}{5} = \frac{12}{5} = \mathbf{2\frac{2}{5} \text{ in}} = \mathbf{2.4 \text{ in}}$.

20. Let x represent the amount of milk needed in the recipe if only 1 cup of flour is used.
$\frac{1\frac{1}{2} \text{ cups flour}}{4 \text{ cups milk}} = \frac{1 \text{ cup flour}}{x \text{ cups milk}} \Rightarrow 1\frac{1}{2}x = 1 \cdot 4 \Rightarrow \frac{3}{2}x = 4 \Rightarrow x = 4 \cdot \frac{2}{3} = \frac{8}{3} = \mathbf{2\frac{2}{3} \text{ cups}}$.

Section 7-1B

2. **(a)** $5000 + 200 + 0.4 = \mathbf{5200.4}$.
(b) $0.004 + 20{,}000 = \mathbf{20{,}000.004}$.
(c) $200 + 30{,}000 = \mathbf{30{,}200}$.
(d) $10^{-3} + 10^{-5} = 0.001 + 0.00001 = 0.00101$.

4. **(a)** $0.45 =$ **forty-five hundredths**.
(b) $2.035 =$ **two and thirty-five thousandths**.
(c) $45.0006 =$ **forty-five and six ten-thousandths**. Six ten-thousandths $= 6 \cdot 10^{-4}$.
(d) $0.0000445 =$ **four hundred forty-five ten-millionths**. Four hundred forty five ten-millionths $= 4 \cdot 10^{-5} + 4 \cdot 10^{-6} + 5 \cdot 10^{-7}$.

6. A rational number in simplest form can be written as a terminating decimal if and only if the prime factorization of the denominator contains no primes other than 2 or 5:

   (a) **Terminating**. The denominator in reduced form contains only 2 as a prime factor.

   (b) **Terminating**. The denominator contains only 2 as a prime factor.

   (c) **Terminating**. The denominator contains only 5 as a prime factor.

   (d) **Nonterminating**. The denominator contains 17 as a prime factor.

   (e) **Terminating**. The denominator contains only 5 as a prime factor.

   (f) **Terminating**. $\frac{14}{35}$ reduces to $\frac{2}{5}$. Since the reduced fraction has only a five in its denominator, the fraction terminates.

11. (a) $\frac{1}{16} = \frac{1}{2^4} = \frac{5^4}{2^4 \cdot 5^4} = \frac{625}{10{,}000} = \mathbf{0.0625}$.

    (b) $224 + \frac{7006}{10{,}000} = 224.7006$.

16. $0.00334_{seven} = \mathbf{0 \cdot 7^1 + 0 \cdot 7^{-1} + 0 \cdot 7^{-2} + 3 \cdot 7^{-3} + 3 \cdot 7^{-4} + 4 \cdot 7^{-5}}$. This is the number in base ten. To remain in base-seven, replace 7 with 10_{seven}.

Section 7-2B

3. There would be a total of $30 + 20 + 10 =$ 60 pounds of nuts; at an average price per pound of \$4.50 he would pay a total of $60 \cdot 4.50 =$ \$270. He has already paid $\$3.00 \cdot 30 + \$5.00 \cdot 20 =$ \$190.00, so he has $\$270.00 - 190.00 =$ **\$80.00** to pay for 10 pounds.

8. $18.5 \text{ cups} = 18.5 \text{ cups} \cdot \frac{1 \text{ quarts}}{4 \text{ cups}} = \frac{185}{40} \text{ quarts} = \frac{37}{8} \text{ quarts} = \mathbf{4.625 \text{ quarts}}$.

18. **(a)** $(5 \cdot 10^7)(7 \cdot 10^{12}) = 5 \cdot 7 \cdot 10^7 \cdot 10^{12}$
$= 35 \cdot 10^{19} = \mathbf{3.5 \cdot 10^{20}}$.

(b) $(\bar{}13 \cdot 10^4) \div 65 = \frac{\bar{}13}{65} \cdot 10^4 =$
$\frac{\bar{}1}{5} \cdot 10^4 = \bar{}0.2 \cdot 10^4 = \mathbf{\bar{}2.0 \cdot 10^3}$.

(c) $(3 \cdot 10^7)(4 \cdot 10^5) \div (6 \cdot 10^{-7}) =$
$\frac{3 \cdot 4}{6} \cdot 10^7 \cdot 10^5 \cdot 10^7 = \mathbf{2.0 \cdot 10^{19}}$.

20. Distance = speed $\times$ time $=$ 55.5 mph $\times$ 0.75 hr $=$ **41.625 miles**.

26. **(a)** $12.4 \cdot 7 = 6.2 \cdot 2 \cdot 7 = 6.2 \cdot \mathbf{14}$.

(b) $12.4 \div 0.2 = \dfrac{12.4}{2 \cdot 0.1} = \dfrac{6.2}{0.1} = \mathbf{6.2} \div 0.1$.

(c) $ab = a \cdot 10^{-1} \cdot \mathbf{10} \cdot \mathbf{6}$.

(d) $12.3 = \dfrac{12.3}{1} = \dfrac{1230}{10^2} = 10^{-2} \cdot \mathbf{1230}$.

Assessment 7-3B

1. In each of the following, divide the numerator by the denominator either with a calculator or by use of the division algorithm. The overline in the quotient indicates that the block of digits underneath is repeated an infinite number of times.

(a) $\frac{2}{3} = 2 \div 3 = \mathbf{0.\overline{6}}$.

(b) $\frac{7}{9} = 7 \div 9 = \mathbf{0.\overline{7}}$.

(c) $\frac{1}{24} = 1 \div 24 = \mathbf{0.041\overline{6}}$.

2. **(a)** $n = 0.\overline{7}$; a one-digit repetend:

$$\begin{array}{rcl} 10n & = & 7.7777\ldots \\ \bar{}n & = & \bar{}0.7777\ldots \\ \hline 9n & = & 7 \end{array}$$

$$\Rightarrow n = \frac{7}{9}.$$

(c) $n = 2.\overline{37}$; a two-digit repetend:

$$\begin{array}{rcl} 100n & = & 237.3737\ldots \\ \underline{\quad {}^{-}n} & = & \underline{\quad {}^{-}2.3737\ldots} \\ 99n & = & 235 \end{array}$$

$$\Rightarrow n = \frac{235}{99}.$$

(e) $n = {}^{-}4.3\overline{4}$; a one-digit repetend:

$$\begin{array}{rcl} 100n & = & {}^{-}434.4444\ldots \\ \underline{\quad {}^{-}10n} & = & \underline{\quad 43.4444\ldots} \\ 90n & = & {}^{-}391 \end{array}$$

$$\Rightarrow n = \frac{{}^{-}391}{90}.$$

11. To find a number halfway between any two others, find their average; i.e., add the original numbers and divide by 2. (Note that $0.\overline{9} = 1.0$.)

$(0.\overline{9} + 1.1) \div 2 = (2.1) \div 2 = \mathbf{1.05}$.

16. Observe that when the repetend is immediatly to the right of the decimal point in the repeating decimal, a in the fraction $\frac{a}{b}$ will be the digit(s) in the repetend and b will be one or more 9's. The number of 9's will be the same as the number of digits in the repetend.

(a) $0.\overline{4} = \frac{4}{9}$.

(b) $0.\overline{12} = \frac{12}{99} = \frac{4}{33}$.

(c) $0.\overline{111} = \frac{111}{999} = \frac{1}{9}$.

Section 7-4B

2. **(a)** $4\frac{1}{2}\% = 4.5\% = \frac{4.5}{100} = \frac{45}{1000} = \mathbf{0.045}$.

(b) $\frac{2}{7}\% = 0.\overline{285714}\% = \frac{0.\overline{285714}}{100}$
$= \mathbf{0.00\overline{285714}}$.

(c) $125\% = \frac{125}{100} = \mathbf{1.25}$.

(d) $\frac{1}{4}\% = 0.25\% = \frac{0.25}{100} = \mathbf{0.0025}$.

3. **(a)** $\underline{\mathbf{5}}$ for every 100. $5\% = \frac{5}{100}$.

(b) $\underline{\mathbf{2.5}}$ for every 50. $5\% = \frac{5}{100} = \frac{2.5}{50}$.

(c) 1 for every $\underline{\mathbf{20}}$. $5\% = \frac{5}{100} = \frac{1}{20}$.

(d) 8 for every $\underline{\mathbf{160}}$. $5\% = \frac{5}{100} \cdot \frac{1.6}{1.6} = \frac{8}{160}$.

(e) 0.5 for every $\underline{\mathbf{10}}$. $5\% = \frac{5}{100} = \frac{0.5}{10}$.

4. **(a)** 63 is 30% of $n \Rightarrow 63 = \frac{30}{100} \cdot n \Rightarrow 30n$
$= 6300 \Rightarrow n = \frac{6300}{30} = \mathbf{210}$.

(b) 7% of $150 \Rightarrow \frac{7}{100} \cdot 150 = \frac{1050}{100} = \mathbf{10.5}$.

(c) 61.5 is n% of $20.5 \Rightarrow 61.5 = \frac{n}{100} \cdot 20.5$
$\Rightarrow 20.5n = 6150 \Rightarrow n = \frac{6150}{20.5} = 300$,
or 61.5 is **300%** of 20.5.

(d) 16 is 40% of $n \Rightarrow 16 = \frac{40}{100} \cdot n \Rightarrow 40n =$
$1600 \Rightarrow n = \frac{1600}{40} = \mathbf{40}$.

18. Techniques may vary:

(a) 10% of $\$42 = \$4.20$. 5% of \$42 is half of 10%, or \$2.10. $\$4.20 + \$2.10 = \mathbf{\$6.30}$.

(b) 10% of $\$280 = \$28$. 20% is twice 10%, or twice $\$28 = \mathbf{\$56}$.

(c) 10% of $\$28 = \$2.80$. 5% is half of 10%, or half of $\$2.80 = \mathbf{\$1.40}$.

(d) 25% is $\frac{1}{4}$; $\frac{1}{4}$ of $\$84 = 84 \div 4$, or **\$21**.

28. Interest charged was $\$28,500 - \$25,000 =$ \$3500. If $I = Prt$ then $r = \frac{I}{Pt} = \frac{\$3500}{\$25{,}000.4} =$
$0.035 = \mathbf{3.5\%}$.

Assessment 8-2B

1. **(a)** If n is the number, then 10 more than the number is $n + 10$.

(b) If n is the number, then 10 less than the number is $n - 10$.

(c) If n is the number, then 10 times it is $10n$.

(d) If n is the number, then the sum of it and 10 is $n + 10$.

(e) If n is the number, its square is n^2 and the difference between its square and itself is $n^2 - n$.

4. **(a)** Cost is $\$(30 + xh)$, where x is the cost per hour for h hours.

(b) Let n, d, and q be the number of nickels, dimes, and quarters, respectively. It is given that $n = 4d$ and $q = 2n = 8d$. Then the value of dimes + value of nickels + value of quarters is $10d + 5(4d) + 25(8d) = 230d$¢.

(c) The sum of the three numbers is $x + (x - 1) + (x - 2) = 3x - 3$.

(d) Make a table of the number of bacteria in terms of n minutes:

n minutes	Number of bacteria
$\frac{1}{2}$	$q \cdot 3$
1	$(q \cdot 3) \cdot 3 = q \cdot 3^2$
$1\frac{1}{2}$	$(q \cdot 3^2) \cdot 3 = q \cdot 3^3$
2	$(q \cdot 3)^3 \cdot 3 = q \cdot 3^4$
⋮	⋮
n	$q \cdot 3^{2n}$

(e) The temperature t hours ago was $(40 + 3t)^\circ$F.

(f) Pawel's salary is $\$s$ the first year; $\$(s + 5000)$ the second year; and $\$(2s)$ the third year, for a total of $\$(4s + 5{,}000)$.

(g) The sum of the three numbers is $x + (x - 2) + (x - 4) = 3x - 6$.

11. **(a)** **(*i*)** The pattern appears to subtract 4 from each term to obtain its successor. The 100th term is given by $3 - 4(99) = -393$.

(*ii*) The nth term is given by $\mathbf{3 - 4(n - 1) = 3 - 4n + 4 = {}^{-}4n + 7}$.

(b) (*i*) The pattern appears to subtract $\sqrt{2}$ from each term to obtain its successor. The 100th term is given by $1 - \sqrt{2}(99) = \mathbf{1 - 99\sqrt{2}}$

(*ii*) The nth term is given by $\mathbf{1 - \sqrt{2}(n - 1)}$ $= 1 - \sqrt{2}n + \sqrt{2} = {}^{-}\sqrt{2}n + 1 + \sqrt{2}$.

(c) (*i*) The pattern appears to add π to each term to obtain its successor. The 100th term is given by $0.5 + \pi + \pi(99) = \mathbf{0.5 + 100\pi}$.

(*ii*) The nth term is given by $0.5 + \pi + \pi(n - 1) = \mathbf{0.5 + \pi n}$.

Section 8-3B

3. Let m be the number of matchsticks Ryan uses. Then $m_1 = 4, m_2 = 7, m_3 = 10, \ldots$. Assume an arithmetic sequence with $d = 3$. Let m_n be the number of matchsticks in the last figure; then $m_n + (m_n - 3) = 599 \Rightarrow m_n = 301$ matchsticks. Thus the last two figures used **298** and **301** matchsticks, respectively.

5. Let e be the amount left to the eldest, m be the amount left to each of two middle siblings, and y be the amount left to the youngest.

It is given that $e = 2y$ and $m = y + 16,000$.

Thus if $e + m + m + y = 1,000,000$ then

$2y + 2(y + 16,000) + y = 1,000,000 \Rightarrow$

$5y + 32,000 = 1,000,000 \Rightarrow y = 193,600.$

So $e = 2y = \mathbf{\$387,200}$

$m = y + 16,000 = \mathbf{\$209,600}$

$y = \mathbf{\$193,600}$.

10. Let a be the length of the parallel side, so the length of each of the perpendicular sides is $2a$.

Then $2a + a + 2a = 800 \Rightarrow a = 160$. The pasture's dimensions are thus **320 by 160 yards.**

Section 8-4B

1. Answers may vary; one such could be 0.454455444555…, and continuing the pattern of adding a 4 and a 5 to each succeeding group.

4. **(a)** **Irrational**. There is no rational number n such that $n^2 = 78$.

(b) **Rational**. $9^2 = 81$.

(c) **Rational**. $7^3 = 343$.

(d) **Rational**. The sum of two rational numbers is rational, since the rationals are closed under addition.

(e) **Irrational**. The quotient of any non-zero rational number and any irrational number is irrational.

10. **(a)** $Q \cap I = \boldsymbol{I}. I \subset Q,$ since the set of rational numbers contains the set of integers.

(b) $S - Q = \boldsymbol{S}.$ There are no irrational numbers which are also rational.

(c) $R \cup S = \boldsymbol{R}.$ $S \subset R,$ or irrational numbers are real.

(d) $\boldsymbol{R}.$ $Q, I, W,$ and S are all subsets of the set of real numbers.

(e) $\overline{S} = \boldsymbol{Q},$ where Q is the set of rational numbers. $Q \cup S = R; Q \cap S = \emptyset.$

11.

		N	I	Q	R	S
(a)	$\sqrt{3}$				√	√
(b)	$4\frac{1}{2}$			√	√	
(c)	$^{-}3\frac{1}{7}$			√	√	

17. $h^2 = 5^2 + 12^2 = 169 \Rightarrow h = \sqrt{169} = \mathbf{13}.$